

ROXSTAR

Voice Draft, Room & Spin Wheel System

Unified Technical Assessment for Audio, Backend and DevOps Candidates

Assessment objective: Build an Android voice-draft feature using Oboe, a Node.js real-time room service, and a multiplayer spin wheel that demonstrates logical reasoning, state management, concurrency handling and cloud deployment.

Assessment detail	Expectation
Total score	200 points
Recommended effort	12-16 focused hours
Submission window	Within 5 calendar days, unless otherwise agreed
Implementation	Individual take-home project
Primary output	Private Git repository, cloud endpoint and 5-10 minute demonstration
Review	Code walkthrough, architecture discussion and reasoning interview

Candidate Name: _____

Role Applied For: Audio / Android | Backend | DevOps

1. Assessment Overview

The solution contains three connected product capabilities: local voice recording and effects, real-time room participation, and a spin wheel played by users inside a room. Live audio streaming, WebRTC, LiveKit, AI, Kubernetes and multi-tenant architecture are outside the scope.

Expected user journey

1. Record a local voice clip on Android using Oboe.
2. Apply Echo, Reverb or Pitch Shift and save the result as a Draft.
3. Create or join a room through the Node.js backend.
4. See participant updates through WebSocket/Socket.IO events.
5. Share a Draft with the room.
6. Start a Spin Wheel when the room has enough eligible users.
7. Receive elimination events until one winner remains.
8. Retrieve the final room and spin result from the backend.

Important: The spin wheel uses virtual points only. No payment, wallet or real-money implementation is required.

Score summary

Section	Maximum
A. Android Audio Studio using Oboe	40
B. Room and Real-Time Communication	40
C. Spin Wheel Logic and Reasoning	50
D. Backend and Database Engineering	30
E. Cloud and DevOps	20
F. Documentation and Communication	20
Total	200

Overall performance levels

Score	Level	Interpretation
170-200	Level 5 - Exceptional	Strong implementation, excellent reasoning, robust state handling and production-minded decisions.
140-169	Level 4 - Strong	Complete core solution with sound design and only minor gaps.
110-139	Level 3 - Meets Expectations	Working core flow with clear reasoning, acceptable quality and documented limitations.
80-109	Level 2 - Developing	Partial implementation; fundamentals are visible but important reliability or design gaps remain.
0-79	Level 1 - Insufficient Evidence	Core flows are incomplete or the candidate cannot explain major implementation decisions.

2. Technology Guide

Technology	Used for	Purpose
Oboe	Android microphone capture, playback and voice-effect processing	High-performance native Android audio inside the local recording pipeline.
WebSocket / Socket.IO	Client-to-server room and spin events	Presence, room state, draft sharing, spin progress, elimination and winner notifications. It does not carry live audio.
Node.js	REST APIs and real-time service	Owns room state, spin state, validation, persistence and event broadcasting.
Database	Users, rooms, drafts, spins and results	PostgreSQL, MySQL or MongoDB may be used. Justify schema, keys and indexes.
Docker	Backend packaging	Repeatable backend build and deployment package.
AWS / GCP / Azure	Final hosting	Choose any one cloud provider. Local-only final deployment is not accepted.

3. Section A - Android Audio Studio using Oboe (40 Points)

A1. Voice recording - 10 points

- Record microphone input using Oboe.
- Support Start, Stop and Cancel.
- Save a valid local audio file.
- Handle microphone permission and expected failures.

Scoring item	Points
Oboe integration and stream lifecycle	5
Valid recording and playback quality	3
Permission and failure handling	2

A2. Draft management - 10 points

- Save recordings as Drafts.
- List Drafts with name, creation time and duration.
- Play and delete a Draft.

Scoring item	Points
Draft save/list/play/delete correctness	5
Android structure and separation of concerns	3
Usability and state presentation	2

A3. Voice effect - 20 points

Implement at least one effect: Echo, Reverb or Pitch Shift. The effect should be part of the Oboe/native audio path where practical.

Scoring item	Points
Working effect implementation	10
Audio processing and buffer design	5
Stability across repeated operations and lifecycle changes	5

Microphone -> Oboe input stream -> Effect processing
-> Encoding / file writer -> Local Draft storage -> Playback

4. Section B - Room and Real-Time Communication (40 Points)

B1. Room management - 20 points

- Create Room
- Join Room
- Leave Room
- Get Room Details and participant list
- Share a selected Draft with the room

Scoring item	Points
REST API design	5
Authoritative room state	5
Validation and invalid-operation handling	5
Room and membership data model	5

B2. WebSocket events - 20 points

Mandatory event	Expected behavior
user_joined	Broadcast updated participant information.
user_left	Broadcast departure and clean presence after leave/disconnect.
draft_shared	Notify members that a Draft has been shared.
spin_started	Publish active spin, eligible players and initial sequence state.
user_eliminated	Publish each elimination with updated remaining players.
winner_announced	Publish final winner and completed spin state.
room_state	Return latest state after connection or reconnection.

Scoring item	Points
Connection and disconnect management	5
Correct event handling and room broadcasting	5
Reconnect logic	5
State synchronization	5

5. Section C - Spin Wheel Logic and Reasoning (50 Points)

C1. Core rules

- Minimum 3 and maximum 20 eligible users per spin.
- An admin or room owner starts the spin manually.
- Only one active spin may exist in a room.
- After start, eliminate one active participant every 5 seconds.
- The last remaining participant is the winner.
- Persist the result and broadcast the complete event sequence.
- Virtual points may be awarded to the winner, but wallet/payment logic is not required.

WAITING -> RUNNING -> COMPLETED
 \-> ABORTED

C2. Basic functionality - 20 points

Scoring item	Points
Validate and start the wheel	5
Generate and process eliminations	5
Select exactly one valid winner	5
Emit events in the correct order	5

C3. State management - 15 points

Scoring item	Points
Room state consistency	5
Spin lifecycle and valid transitions	5
Participant eligibility and status tracking	5

C4. Edge-case reasoning - 15 points

Identify, implement and document edge cases. Suggested cases include duplicate start requests, simultaneous joins, user departure during a spin, reconnect during a spin, admin disconnect, insufficient players, last players leaving, duplicate events, delayed timers and server restart.

Implemented and demonstrated	Points
1-2 meaningful edge cases	5
3-4 meaningful edge cases	10
5 or more meaningful edge cases	15

Quality rule: Edge cases count only when the behavior is implemented or convincingly tested, and the candidate can explain the chosen outcome.

6. Section D - Backend and Database Engineering (30 Points)

D1. Required APIs and services

- Create Room
- Join Room
- Leave Room
- Get Room State
- Share Draft
- Start Spin
- Get Spin State or Result
- Health / readiness endpoint

D2. Database design - 15 points

Suggested entity	Purpose
User	Participant identity and profile metadata.
Room	Room owner, status and timestamps.
RoomMember	Membership and connection state.
Draft	Recording metadata and hosted file location.
Spin	Room, status, start/completion time and winner.
SpinParticipant	Eligibility, elimination order/time and final status.
SpinEvent / Result	Auditable event or final outcome record.

Scoring item	Points
Schema and keys	5
Relationships, constraints and indexes	5
Queries and persistence correctness	5

D3. Backend quality - 15 points

Scoring item	Points
Validation, error handling and idempotency	5
API/service organization and logging	5
Unit and integration testing	5

7. Section E - Cloud and DevOps (20 Points)

- Package the Node.js backend using Docker.
- Provide a CI/CD pipeline that installs dependencies, runs tests, builds and deploys.
- Deploy to AWS, Google Cloud Platform or Microsoft Azure.
- Provide a working hosted endpoint or clear deployment evidence.
- Use environment-based configuration and document secrets handling.
- Document health verification and rollback approach.

Scoring item	Points
Docker packaging	5
CI/CD automation	5
Working cloud deployment	5
Environment configuration and release safety	5

Cloud requirement: Docker may be used locally during development, but a local-only backend is not accepted as the final submission.

8. Section F - Documentation and Communication (20 Points)

Deliverable	Points
README with clean setup, run, test and deployment instructions	5
System architecture diagram	4
Audio-flow diagram	3
Room and WebSocket event-flow diagram	3
Spin state-machine / sequence diagram	3
Assumptions, edge cases, trade-offs and known limitations	2

Required submission package

- Private Git repository with meaningful commits.
- Android app and native Oboe code.
- Node.js backend, database schema/migrations and tests.
- Dockerfile, CI/CD configuration and cloud deployment assets.
- API documentation using OpenAPI/Swagger or an equivalent format.
- 5-10 minute demo recording covering audio, room, spin wheel and cloud deployment.

Recommended repository structure

```

/android-app
/native-audio
/backend
/database
/infrastructure
/docs/architecture
/docs/api
/tests
README.md

```

9. Demonstration Checklist

- Record a voice clip using the Oboe path.
- Apply Echo, Reverb or Pitch Shift.
- Save, list, play and delete Drafts.
- Create a room and join from a second client.
- Show user_joined, user_left and draft_shared events.
- Start a spin with at least 3 users.
- Show elimination events every 5 seconds and exactly one winner.
- Demonstrate one reconnect or state-recovery flow.
- Demonstrate at least three implemented edge cases.
- Show automated tests, hosted endpoint and CI/CD evidence.
- Trigger one expected failure and explain the handling.

10. Candidate Submission Checklist

- ☐ Private repository access granted
- ☐ README includes clean-start instructions
- ☐ Oboe recording path works
- ☐ One allowed voice effect works
- ☐ Draft lifecycle works
- ☐ Room create/join/leave works
- ☐ Mandatory WebSocket events work
- ☐ Spin lifecycle and winner selection work
- ☐ Edge cases are documented and demonstrated
- ☐ Database diagram and migrations included
- ☐ Architecture, audio, event-flow and spin diagrams included
- ☐ API documentation included
- ☐ Docker and CI/CD included
- ☐ AWS/GCP/Azure deployment evidence included
- ☐ Automated test results included
- ☐ Demo recording link included

Submission details

Field	Candidate entry
Candidate name	
Role applied for	
Repository link	
Demo recording link	
Cloud provider and endpoint	
Android device	
Implemented effect	
Handled edge cases	
Self-assessed score / 200	

Final note: Correctness, reasoning and the ability to explain trade-offs are more important than visual polish or feature volume.